

## Ejercicio 9

①

$$\begin{array}{c}
 \textcircled{I} \\
 \frac{\frac{\frac{\frac{\frac{\frac{\Gamma' \vdash P}{Ax} \quad \frac{\Gamma' \vdash \neg P}{Ax}}{\Gamma \vdash \perp}{\neg e}}{\vdash e}}{\frac{\Gamma, \neg P, P \vdash Q}{Ax}}{\Gamma \vdash P \Rightarrow Q}{\Rightarrow i}}{\frac{\Gamma, \neg P \vdash P \Rightarrow Q}{\Rightarrow e}}{\frac{\Gamma, \neg P \vdash Q \Rightarrow P}{Ax}}{\Gamma, \neg P \vdash P}{\Rightarrow e}}{\frac{\Gamma, \neg P \vdash \perp}{\neg e}}{\Gamma, \neg P \vdash \neg P}{Ax}}{\Gamma, \neg P \vdash \neg \neg P}{\neg e} \\
 \hline
 \Gamma = ((P \Rightarrow Q) \Rightarrow Q), (Q \Rightarrow P) \vdash P \quad \text{PBC} \\
 \hline
 ((P \Rightarrow Q) \Rightarrow Q) \vdash ((Q \Rightarrow P) \Rightarrow P) \quad \Rightarrow i \\
 \hline
 ((P \Rightarrow Q) \Rightarrow Q) \Rightarrow ((Q \Rightarrow P) \Rightarrow P) \quad \Rightarrow i
 \end{array}$$

II

$$\begin{array}{c}
 \frac{}{\Gamma \vdash P \vee \neg P} \text{LEM} \quad \frac{\frac{Ax}{\Gamma_1 \vdash P \Rightarrow Q} \quad \frac{Ax}{\Gamma_1 \vdash P}}{\Gamma_1 = \Gamma, P \vdash Q} \Rightarrow e \quad \frac{\frac{Ax}{\Gamma_2 \vdash \neg P \Rightarrow Q} \quad \frac{Ax}{\Gamma_2 \vdash \neg P}}{\Gamma_2 = \Gamma, \neg P \vdash Q} \Rightarrow e \\
 \hline
 \Gamma = (P \Rightarrow Q) \quad (\neg P \Rightarrow Q) \vdash Q \quad \vee e \\
 \hline
 (P \Rightarrow Q) \vdash (\neg P \Rightarrow Q) \Rightarrow Q \quad \Rightarrow i \\
 \hline
 (P \Rightarrow Q) \Rightarrow ((\neg P \Rightarrow Q) \Rightarrow Q) \quad \Rightarrow i
 \end{array}$$